THIRTY-DAY BUILD PLAN

Paced for full-time work. Hour estimates are given per task so progress can be discussed in terms of effort rather than vibes — they are estimates for planning, not targets to be hit, and an unfamiliar build will beat some of them and blow through others.

Sequence rationale: shared foundations first, then football (better evidenced, higher volume, launch first), then tennis, then the measurement layers, then a proper walk-forward backtest, then paper trading. Measurement is deliberately built before the models are trusted and after they exist — early enough to catch problems, late enough to have something to measure.

How to use the done-list. Each day has deliverables with a tick box and a blockers line. Fill in blockers honestly — a blocked day with a named cause is more useful than a day that silently slipped. Software slips for real reasons, and the blockers log is what distinguishes a genuine obstacle from a quiet week.

WEEK 1 — FOUNDATIONS

Both repositories. Nothing sport-specific gets built until the plumbing is right.

Day 1 — Environment, repositories, database

Task	Est.
Two repositories created, Python environments, dependency management, linting, pre-commit	2h
PostgreSQL running; both databases created	1h
Core schema written and migrated: `fixtures`, `odds_snapshots`, `ratings`, `predictions`, `decisions`, `settlements`, `clv`, `data_health`	4h
`as_of` and `model_version` columns present on every relevant table (rules R5, R7)	included
README in each repo pointing at the specs	1h

Deliverables

- Two repos, both running a hello-world job against their own database
- Schema migrated, `\dt` shows all tables in both
- `odds_snapshots` is append-only — enforced, not just intended

Blockers:

Day 2 — The arithmetic library

The single most reused code in the project. Get it right now and never revisit it.

Task	Est.
EV, fair odds, break-even probability, booksum, overround, hold	1h
De-vig: multiplicative, additive, **power**, **Shin**, odds-ratio	3h
Root-finding for power and Shin (`scipy.optimize.brentq`)	2h
Kelly, quarter-Kelly with 2% cap	1h
CLV **including the `(1 + closing_margin)` division**	1h
Sample-size calculator $N = z^2(o-1)/E^2$	0.5h
Unit tests: 2-way, 3-way, and a longshot book (1.25 / 4.50 / 15.00)	2h

Deliverables

- Power and Shin both converge to $\Sigma p = 1$ within 1e-9 on all test books
- Longshot test reproduces the expected divergence — the 15.00 leg's fair price should vary substantially by method while the 1.25 leg barely moves
- Every de-vig function returns the method name alongside the probabilities
- Library copied into both repos (copied, not shared — the systems stay decoupled)

Blockers:

Day 3 — Odds feed and the sharp reference problem

This is the day the project's biggest external dependency gets resolved. Do not defer it.

Task	Est.
Evaluate odds providers against requirements: book coverage, sharp reference availability, call limits, historical access	3h
Resolve sharp reference access. Pinnacle does not accept UK customers; determine whether the chosen provider carries their prices as data, or fall back to Betfair Exchange	3h
API keys obtained, stored in a secrets manager — never in the repository	1h
Odds ingest job written: polls, stores append-only with `observed_at`	2h

Deliverables

- Named sharp reference source, with its measured booksum on real fixtures (not an assumed margin)
- Exchange commission conversion implemented if Betfair is used: `o_eff = 1 + (o−1)(1−c)`
- First odds snapshots landing in the database
- Rate limits documented; ingest respects them

Blockers:

Day 4 — Football historical data

Task	Est.
football-data.co.uk full archive downloaded — all in-scope leagues, all available seasons	2h
Loader written; idempotent re-runs	3h
Team name normalisation table (this will be worse than expected)	2h
Data quality report: missing shots/corners by league-season, coverage gaps	2h

Deliverables

- Full archive loaded with `as_of` populated
- Coverage report showing exactly which league-seasons have shots and corners, and which do not
- Confirmed: **no xG for League One and League Two** — design accounts for it

Blockers:

Day 5 — Tennis historical data

Task	Est.
Sackmann `tennis_atp` cloned, all match files loaded including Challenger	2h
tennis-data.co.uk odds archive loaded	2h
Player name normalisation between the two sources — budget properly, this is the painful one	3h
Data quality report: serve/return stat availability by tour level and year	2h

Deliverables

- Match history and odds joined, with a measured join rate reported per year
- Coverage report by tour level — expect Challenger match-level stats to be present and point-by-point charting to be thin
- Unmatched names logged, not silently dropped

Blockers:

WEEK 2 — FOOTBALL

Day 6 — GAP ratings engine, part one

Task	Est.
Read Wheatcroft (2020) properly, not just the spec	2h

Rating state model: attacking and defensive, home and away separately, per statistic	3h
Update mechanic implemented for a single statistic (shots)	3h
Hand-computed worked example to verify the update arithmetic	1h

Deliverables

- Shots GAP ratings computing across one league-season
- Update verified against a hand-computed example
- **Goals are not an input anywhere** (rule R1) — assert this in a test

Blockers:

Day 7 — GAP ratings engine, part two

Task	Est.
Extend to shots on target and corners as independent rating systems	3h
Learning rate λ fitted by walk-forward, per statistic per league	3h
Variant counter: every fitted configuration logged	1h
Ratings persisted with `as_of` and `model_version`	2h

Deliverables

- Three independent GAP systems running across all in-scope leagues
- Fitted λ values with the walk-forward evidence behind them
- Variant count recorded for every fit

Blockers:

Day 8 — Statistics to goal expectation

Task	Est.
Shot-to-goal conversion model, fitted per league	3h

GAP-predicted statistics → expected goals per side	2h
Corner-count distribution: test Poisson against negative binomial	3h

Deliverables

- Expected goals per side derived from predicted statistics, not from historical goals
- Corner distribution chosen with the comparison evidence recorded
- Sanity check: predicted totals across a season roughly match observed totals

Blockers:

Day 9 — Scoreline model

Task	Est.
Dixon-Coles: $\lambda_home = \exp(\alpha_home + \beta_away + \gamma)$, $\lambda_away = \exp(\alpha_away + \beta_home)$	3h
Tau low-score correction with ρ fitted	3h
Time decay implemented — and confirmed to buy almost nothing	1h
Full 2-D scoreline matrix, $i, j = 0 \ldots 10$	2h

Deliverables

- Fitted γ in the region of 0.29 on a Premier League season; ρ around −0.13
- **Unit test: $\rho = 0$ collapses to plain Poisson**
- Test confirming the tau correction raises draw probability
- Time-decay result documented (expect optimal $\xi \approx 0$ within a season)

Blockers:

Day 10 — Matrix to market prices

Task	Est.

1X2 from triangles and diagonal	1h
Over/under any line	1h
Both teams to score	1h
Asian handicap by `i − j`, quarter-lines split into half-stakes, integer lines as push	3h
Corner totals from the corner distribution	2h
Cross-check: AH and 1X2 must agree — a disagreement is a bug	1h

Deliverables

- All in-scope markets priced from one scoreline object
- AH/1X2 consistency test passing
- Quarter-line handling verified against a worked example

Blockers:

Day 11 — Football manager and thresholds

Task	Est.
Slate generator: every fixture in scope, every matchday, **no filtering by attractiveness**	2h
Reference price de-vig with power/Shin	2h
Edge calculation and threshold table by competition group	2h
Cup restriction enforced in code: corners and BTTS only, hard constraint	2h
Minimum acceptable price on every selection	1h

Deliverables

- Slate produces every in-scope fixture; count matches the fixture list exactly
- Thresholds: 5% top leagues, 6% English lower, 8% Scotland, 8% cups
- **Test proving a 1X2 selection on a cup tie is rejected**

- Decisions logged with reason codes, including no-bets

Blockers:

Day 12 — Asian handicap line-type rule, xG check

Task	Est.
Line-type preference: integer → .75 → .25 → .5	2h
Line type recorded on every pick record	1h
xG ingest for top-five leagues	3h
xG sanity check as a **veto input only**, never a rating input	2h

Deliverables

- Line-type rule applied and auditable in the pick records
- xG check running where data exists, skipping League One and League Two cleanly
- Test asserting xG never reaches the GAP ratings

Blockers:

WEEK 3 — TENNIS

Day 13 — Surface Elo

Task	Est.
Four ratings per player: overall, hard, clay, grass	2h
Standard Elo update with decaying K by career match count	3h
50/50 surface blend	1h
K-parameters fitted by Brier minimisation, walk-forward, variants counted	3h

Deliverables

- Elo running across the full Sackmann history including Challengers
- Blend fixed at 50/50; any alternative tested is counted
- Brier score reported — **expect around .180–.190, worse than the market's .172**

Blockers:

Day 14 — Serve/return statistics and opponent adjustment

Task	Est.
SPW and RPW per player per match from Sackmann data	2h
Trailing-window aggregation with surface weighting and a minimum-match threshold	2h
Shrinkage toward the surface tour average below the threshold	1h
Opponent adjustment: `p_A_serve = SPW_A - (RPW_B - RPW_tour_avg)`	2h
Sum constraint: rescale so `p_A + p_B = t_surface`	2h

Deliverables

- Adjusted serve probabilities produced for any fixture
- Sanity check against benchmarks: top-100 ≈ 80% service games held, 20% return games won
- **Test asserting neither player's serve probability can exceed a plausible ceiling**

Blockers:

Day 15 — Hierarchical Markov propagation

Task	Est.
Game recursion `N(a,b) = p·N(a+1,b) + (1−p)·N(a,b+1)` with deuce handling	3h
Tiebreak recursion with alternating serve	2h
Set recursion	2h

Match: best-of-three and best-of-five, plus per-event final-set rules	2h
Memoisation	1h

Deliverables

- Exact match probability from two serve numbers, no simulation needed
- Verified against a published worked example
- **Final-set rules correct per event per year** — these changed within the backtest window
- Format check: best-of-five gives the favourite a higher probability than best-of-three at the same serve inputs

Blockers:

Day 16 — Common-opponent routing

Task	Est.
Shared-opponent set computation per fixture	3h
Serve/return performance restricted to the common opponent set	3h
Fallback to global tour average where the shared set is too thin	1h
Integration as an **input to `p_serve` only**	1h

Deliverables

- Common-opponent routing improving `p_serve` estimates at Challenger level, where players share opponents constantly
- **Test asserting common-opponent output cannot fire a bet on its own and cannot relax any threshold** — the published +3.8% result failed replication and this constraint is why
- Minimum shared-opponent count configured and enforced

Blockers:

Day 17 — Intransitivity graph

Task	Est.
Read arXiv 2510.20454	2h
Directed weighted graph: nodes = players, edges loser → winner, recency-decayed	3h
Cycle detection and **intransitivity score** per fixture	3h

Deliverables

- Intransitivity score computed for any fixture
- **Build the filter before the model** — the score is the valuable object
- Distribution of scores across a season inspected; threshold for "high intransitivity" set from that distribution, not guessed

Blockers:

Day 18 — Intransitivity model and abstention

Task	Est.
Model over the graph structure producing a match probability	4h
Abstention on low-intransitivity fixtures as a distinct output type	2h
Walk-forward check: applied indiscriminately, does it lose?	2h

Deliverables

- Agent abstains on the large majority of fixtures — this is correct behaviour
- Indiscriminate application reproduces a clear loss (published figure: –5.49%). **If it doesn't, the filter is not doing what you think it is**
- Selective application evaluated on its own, with the bet count reported

Blockers:

Day 19 — Tennis manager

Task	Est.
Process isolation: the three agents cannot see each other's outputs or state	3h
Consensus rules C1–C4: direction, 8-point spread ceiling, 50% floor, abstention handling	3h
De-vig, edge calculation, threshold table	2h

Deliverables

- Blindness enforced architecturally, not by convention — demonstrate it
- Thresholds: 5% tour, 8% Challenger, 8% anything with a leg above 4.00
- Every consensus failure logged with which rule rejected it
- Minimum acceptable price on every selection

Blockers:

WEEK 4 — MEASUREMENT, SAFETY, VALIDATION

Day 20 — Price scanners

Task	Est.
Multi-book polling for both systems	3h
Best-available price identification per selection per market	2h
Type A (model edge) and Type B (price dispersion) as distinct flag types	3h

Deliverables

- Both scanners running, coverage count reported per fixture
- Type A and Type B recorded separately in the decision log — **never mixed**
- Corner and card markets restricted to the tightest-margin book, with booksum recorded

- Coverage expansion listed as ongoing work: more books beats better models

Blockers:

Day 21 — Veto agents

Task	Est.
Tennis: withdrawals, injury reports, previous-match load, turnaround, first match back, schedule changes	4h
Football: team news, lineups, congestion flag, weather, referee for cards, postponements	4h

Deliverables

- Both agents **kill only** — architecturally incapable of creating a selection or altering a probability
- Neither agent can read the market price
- Every veto logged with its trigger, so veto value can be measured later

Blockers:

Day 22 — Freshness gate, monitoring, kill switch

Task	Est.
Freshness gate: odds ≤ 30 min, results ≤ 24h, team news ≤ 6h; **blocks publication**	3h
Alerting: CLV drift (14 days negative), calibration deviation, zero-volume, job failure	3h
Kill switch: CLV stop at 200 settled selections, plus drawdown stop	2h

Deliverables

- **Freshness gate tested by deliberately staling a feed** — publication must halt
- No silent fallback to cached data anywhere
- **Kill switch tested by injecting a synthetic losing run**

- Drawdown number agreed and written down before launch, not after

Blockers:

Day 23 — Calibration and CLV pipelines

Task	Est.
Closing price capture at event start, automated, both systems	3h
CLV computation **including the margin division**	2h
Calibration: decile binning, reliability diagram, Brier by market and model version	3h

Deliverables

- Closing prices captured automatically — no manual step
- CLV formula verified against a hand-computed example
- Reliability diagrams rendering for both sports
- Brier scores reported against the market's as a reference point

Blockers:

Day 24 — Scheduling and publication

Task	Est.
Scheduler configured, all jobs, both systems	3h
Tennis: 22:00 recompute, 00:00 publish, 07:00 withdrawal sweep, 12:00 second sweep	2h
Football: Monday internal slate, 08:00 match-day publish, kick-off minus 3h retraction sweep	2h
UTC storage with per-tournament schedules derived backwards from first serve	2h

Deliverables

- Full schedule running end to end
- Non-European tennis tournaments scheduled correctly, not on a fixed UK clock
- **Empty slate handled cleanly** — the system can publish “nothing today”
- Retraction path tested

Blockers:

Day 25 — Football walk-forward backtest

Task	Est.
Harness: train years 1–N, test N+1, roll	3h
Assertion that no feature’s `as_of` post-dates its fixture	2h
Opening odds for the strategy, closing odds for validation	2h
Full run across the archive	2h

Deliverables

- Look-ahead assertion in place and failing loudly when violated
- Results reported in order: **calibration, then CLV, then ROI**
- Variant count reported alongside every result
- Expect around +0.8% per bet at best, realistically less. **A backtest showing 5% is a bug**

Blockers:

Day 26 — Tennis walk-forward backtest

Task	Est.
Same harness applied to tennis	3h
Per-agent evaluation as well as consensus	3h

Type A versus Type B breakdown	2h

Deliverables

- Per-agent Brier scores reported
- **Type B flags substantially outnumbering Type A** — if Type A dominates, the de-vig or the threshold is wrong
- **Outlier check**: if any large ROI rests on a small number of bets, find the outlier before reporting anything. This is exactly how the common-opponent result went wrong

Blockers:

Day 27 — Dashboard

Task	Est.
Reliability diagrams, both sports	2h
CLV over time, cumulative and rolling	2h
Type A / Type B split	1h
Decision log browser including no-bets	2h
Data health, feed freshness, alert history	2h

Deliverables

- Every measurement layer visible in one place
- **Drawdown contextualised** — a long losing run must not look like a broken model. A working system with 6,562 selections (expected 4.19% yield, actual 3.79%) still produced 1,000-bet windows that looked like 1-in-40 failures
- Model version filter on every view

Blockers:

Day 28 — Paper trading launch

Task	Est.
Paper ledger: selections generated, published internally, priced, settled, no money	3h
Suggested stake fraction: **quarter Kelly capped at 2%**	2h
End-to-end dry run on live fixtures, both sports	3h

Deliverables

- Both systems producing live selections into the paper ledger
- Staking rule implemented and recorded on every pick
- **Paper mode is permanent** — live use runs alongside it, never instead of it

Blockers:

Day 29 — Hardening

Task	Est.
Failure-mode testing: API down, malformed response, missing fixture, duplicate ingest	4h
Idempotency verified on every job	2h
Secrets audit — no keys in the repositories	1h
Backup and restore tested	1h

Deliverables

- Every failure mode produces an alert and a halt, never a silent bad number
- Re-running any job produces identical state
- Restore from backup verified, not assumed

Blockers:

Day 30 — Handover

Task	Est.
Runbook: what runs when, what each alert means, what to do about it	3h
Known limitations documented honestly	2h
Variant counts and fitted parameters collected in one place	2h
Walkthrough	2h

Deliverables

- Runbook complete
- **Limitations written down**, including: the football edge is under 1% and has been public since 2020; the tennis model agents are expected to be worse-calibrated than the market; the intransitivity result has not been independently replicated
- Every fitted parameter and every variant count in one document
- Both systems running unattended for 48 hours without intervention

Blockers:

AFTER DAY 30

Do not treat a profitable backtest as a result. Three things must be true before anything is taken seriously:

1. **Calibration is acceptable.** If the model says 60% and it happens 45% of the time, every edge calculation is wrong and the backtest is meaningless.
2. **CLV is positive over at least 50–100 selections.** This is the early signal, and it validates roughly 75× faster than profit does.
3. **The variant count is known.** Ten configurations tested and one looking excellent is approximately what noise produces.

A backtest validates the *idea*. Forward-testing validates *this implementation of it* — same theory, different code, and a dozen small choices any one of which can flip the result. That is what the paper ledger is for, and why it never gets switched off.

Judge early live selections on closing line value, not on whether they won. Beating the closing line during a losing week is the system working. Winning a week while losing to the close is luck, and it will not persist.